

Digital Buck-Boost (LTC3440) — Working Document

Digital Audio Player · Rev 4

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1. Part Identification

Field	Value
Manufacturer / MPN	Analog Devices — LTC3440EMS#TRPBF
Package	10-lead MSOP (MS)
Function	Synchronous buck-boost, SYS → +3.3 V digital rail

2. Design Requirements

Requirement	Value
Input	SYS, 3.0 – 4.2 V
Output	3.3 V
Load	Digital branch: ESP32-S3, microSD, display logic, DAC digital
Output current	300 mA max ($\approx$ 253 mA S4 peak)
Switching frequency	600 kHz (0.48 – 0.72 MHz), separated from the analog LTC3440 (0.96 – 1.44 MHz)
Inductor ripple target	30%
Mode	Fixed frequency (MODE/SYNC to GND)

Constraints: Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	RT	R_T to GND
2	MODE/SYNC	GND
3	SW1	L
4	SW2	L
5	GND	GND
6	VOUT	+3V3, C_OUT, R1
7	VIN	SYS, C_IN
8	SHDN/SS	VIN (always on, per typical application)
9	FB	R1 / R2 midpoint
10	VC	C_P1 + R_Z to FB

4. External Components

Ref	Value	Rating / package	MPN	AL-PN
L	18 μ H $\pm$ 20%	I _{sat} 1.8 / 2.8 / 3.7 A (10 / 20 / 30% drop), I _{rms} 3.3 A (20 °C rise), 56.5 m Ω typ / 62.5 m Ω max, shielded, 5.48 $\times$ 5.28 $\times$ 5.10 mm	Coilcraft XGL5050-183MEC	#1-094
C_IN	4.7 μ F	16 V X7R 0805	Murata GRM21BR71C475KE51L	#1-071
C_OUT	22 μ F	25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073
R1	172 k Ω 0.1% (E192)	0402	TBD	—
R2	100 k Ω 0.1%	0603	YAGEO RT0603BRD07100KL	#1-066
R_T	100 k Ω 0.1%	0603	YAGEO RT0603BRD07100KL	#1-066
R_Z	15 k Ω 0.1%	0402	YAGEO RT0402BRD0715KL	#1-078
C_P1	1.5 nF $\pm$ 2%	50 V C0G 0402	Murata GRM1555C1H152GA01D	#1-077

5. Equations and Results

Switching frequency

$$R_T = \frac{6 \times 10^{10}}{f} = \frac{6 \times 10^{10}}{600 \times 10^3} = 100 \text{ k}\Omega \quad (0.48 - 0.72 \text{ MHz at } \pm 20\%)$$

Output voltage

$$V_{OUT} = 1.22 \left(1 + \frac{R_1}{R_2} \right) \Rightarrow R_1 = 100 \text{ k}\Omega \left(\frac{3.3}{1.22} - 1 \right) = 170.5 \text{ k}\Omega$$

$$V_{OUT} = 1.22 \left(1 + \frac{172}{100} \right) = 3.318 \text{ V}$$

Inductor (f in MHz, 300 mA, 30% ripple)

$$L_{boost} > \frac{V_{IN(MIN)}(V_{OUT} - V_{IN(MIN)})}{f I_{OUT(MAX)} \text{Ripple } V_{OUT}} = \frac{3.0 \times 0.3}{0.6 \times 0.3 \times 0.3 \times 3.3} = 5.05 \mu\text{H}$$

$$L_{buck} > \frac{V_{OUT}(V_{IN(MAX)} - V_{OUT})}{f I_{OUT(MAX)} \text{Ripple } V_{IN(MAX)}} = \frac{3.3 \times 0.9}{0.6 \times 0.3 \times 0.3 \times 4.2} = 13.1 \mu\text{H} \rightarrow 18 \mu\text{H}$$

Ripple with 18 μH

$$Ripple_{buck} = 30\% \times \frac{13.1}{18} = 21.8\% (65 \text{ mA}_{pp}), \quad Ripple_{boost} = 30\% \times \frac{5.05}{18} = 8.4\% (25 \text{ mA}_{pp})$$

Peak inductor current (3.0 V in, $\eta \approx 0.85$)

$$I_{L,pk} = \frac{0.3 \times 3.3}{3.0 \times 0.85} + \frac{0.025}{2} = 0.40 \text{ A} \quad (4.2 \text{ V in: } 0.3 + 0.033 = 0.33 \text{ A})$$

Inductor copper loss (3.0 V in, 300 mA)

$$P_{DCR} = I_L^2 R_{DCR} = 0.388^2 \times 0.0625 = 9.4 \text{ mW}$$

Output capacitance (0.5% ripple)

$$C_{OUT} \geq \frac{I_{OUT}(V_{OUT} - V_{IN(MIN)}) \cdot 100}{\%Ripple \cdot V_{OUT}^2 \cdot f} = \frac{0.3 \times 0.3 \times 100}{0.5 \times 3.3^2 \times 0.6 \times 10^6} = 2.75 \mu\text{F} \quad (22 \mu\text{F} \text{ fitted})$$

Gate-charge quiescent current (f in MHz)

$$I_Q = f(0.75 V_{IN} + 0.25 V_{OUT}) = 0.6 \times (0.75 \times 4.2 + 0.25 \times 3.3) = 2.385 \text{ mA} \quad (1.85 \text{ mA at } 3.0 \text{ V})$$

Four-switch region

$$V_{IN} = V_{OUT}(1 - 150 \text{ ns} \cdot f) \text{ to } \frac{V_{OUT}}{1 - 150 \text{ ns} \cdot f} = 3.00 \text{ to } 3.63 \text{ V} \quad (\text{buck only above } 3.63 \text{ V})$$

Boost right-half-plane zero (3.0 V in, 300 mA, 18 μH)

$$f_{RHPZ} = \frac{V_{IN}^2}{2\pi I_{OUT} L V_{OUT}} = \frac{3.0^2}{2\pi \times 0.3 \times 18 \mu\text{H} \times 3.3} = 80 \text{ kHz}$$

6. Important Values

Item	Value
Switch current limit	2.7 A typ (2.5 – 2.9 A)
Frequency accuracy	$\pm 20\%$ (0.8 – 1.2 MHz at 1 MHz nominal)
Output	3.318 V with 172 k Ω
Inductor ripple	21.8% buck / 8.4% boost with 18 μH
Peak inductor current	0.40 A at 300 mA, 3.0 V in — well below the 1.8 A 10%-drop point

Item	Value
Inductor at switch limit	2.9 A max $\rightarrow \approx 21\%$ inductance drop (2.8 A at 20%, 3.7 A at 30%); fault only, soft saturation
Inductor copper loss	≈ 9.4 mW at 300 mA
Gate-charge I_Q	2.385 mA at 4.2 V in (≈ 10 mW)
Output capacitance	2.75 μ F required, 22 μ F fitted
Boost RHP zero	≈ 80 kHz — loop crossover must stay well below

7. Open Problems

#	Problem
1	R1 MPN (172 k Ω 0.1% 0402) — add to library
2	V_C network (1.5 nF / 15 k Ω) carried over from the 6.8 μ H / 1.2 MHz design — verify loop with 18 μ H at 600 kHz (load step sim)
3	SHDN/SS tied to VIN — no soft-start; confirm inrush into downstream capacitance
4	Charger (2.7 – 3.3 MHz) overlaps LT3462A (2.0 – 3.5 MHz) while plugged in — risk register

8. Revision Log

Rev	Date	Change
0	2026-09-23	Initial issue: 600 kHz, 3.3 V / 300 mA, inductor, output capacitance, quiescent current.
1	2026-09-23	MODE/SYNC to GND; R1 172 k Ω (3.318 V); inductor $I_{sat} > 3.5$ A.
2	2026-09-23	Inductor Coilcraft XGL5050-183MEC (18 μ H, 3.7 A I_{sat} , 62.5 m Ω); ripple, peak, RHP zero updated.
3	2026-09-23	Inductor I_{sat} by drop column recorded: 3.7 A is the 30% figure; 2.9 A switch limit $\approx 21\%$ drop.
4	2026-09-23	Inductor assigned #1-094.